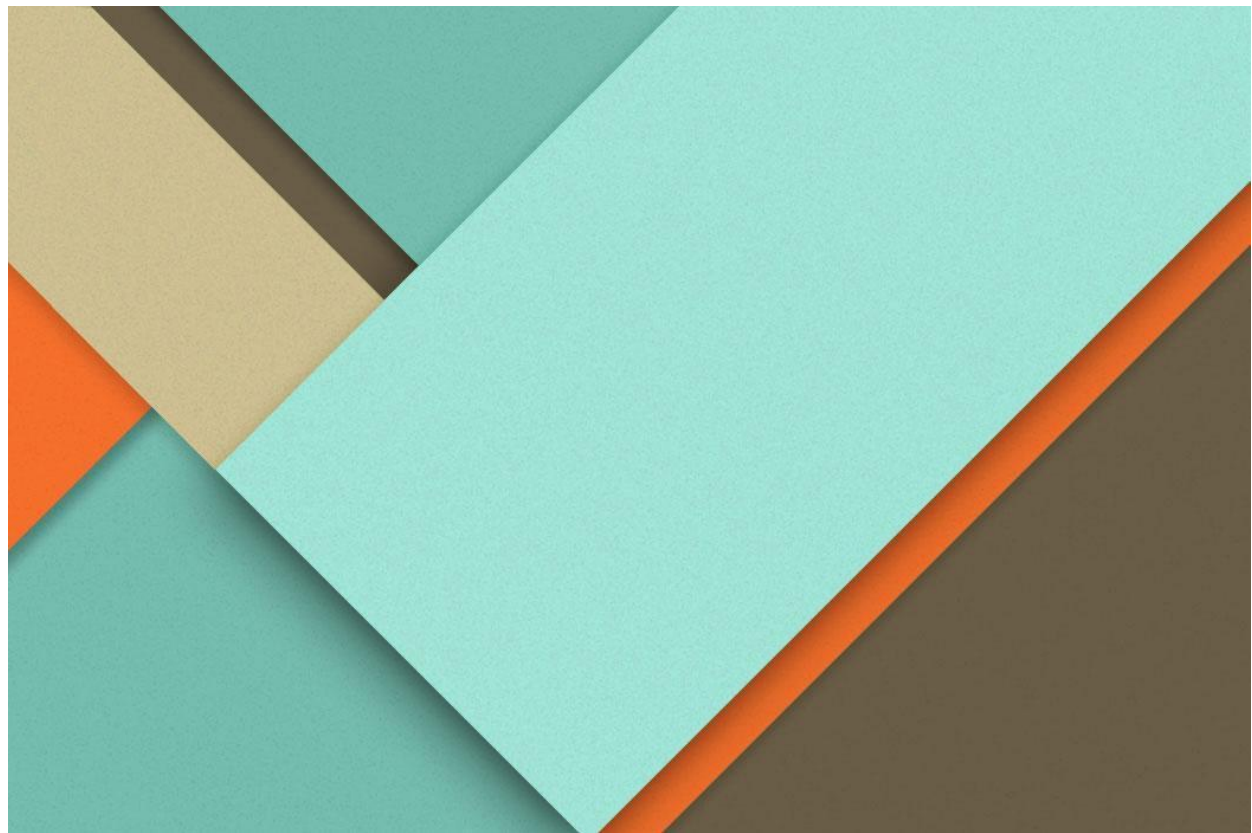




CLINIC'S MANAGEMENT SYSTEM

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Natural language

The management system of a clinic wants to manage its employees, clients, departments and appointments.

The database should keep track of patient details such as name, age, gender, contact details, medical history, and insurance details.

Employees details such as name, contact details, role, and schedules.

Each department is identified by the id; has a name and is used to take appointments.

Appointments details such as patient name, doctor name, appointment date, and time.

An employee has different schedules according to the departments she/he works for.

Person is the “father” of Patient and Employee, so the primary key of Employee and Patient refers to person_id of Person.

Person is identified by an id and not by the Fiscal Code because there could be the case that people have the same Fiscal Code because they have the same name, surname, date of birth and city of birth.

In Patient there is a multiple valued and optional attribute that is allergies, so the Patient can or not have allergies.

The Insurance entity is identified only by the insurance_id.

Appointment is identified by the combination of: date (representing the day of the appointment), the Patient and Employee PK (to see who took the appointment and the doctor that did it) and the Department PK (to see in which department the appointment takes place).

Department is identified by an opposite id and also the Clinic has a personal id that identifies it.

Finally, Schedule represent the time table of an employee.

It is identified by: date (the day in which the employee works), start_time (at what time the employee starts work), the PK of the Department and of the Employee.

I don't want to use an id to uniquely identify the Schedule because it can be identified uniquely also in this case. To compose the primary key, I also use start_time because on the same day, an Employee could work multiple times such as in the morning and in the night.

The relation “work for” between Employee and Department have an attribute: role.

This attribute is used to define the role of the Employee in that Department.



An Employee can have different roles depending on the department they work in.

Through a query on the Appointment entity, it is possible to get the Clinical History of a Patient.

There is also the point, that, since it is a Health Clinic System, there could be different clinics in the same system.

So, since there are more Clinics to manage, Clinic became an entity.

The cardinality constraint between Employee-Schedule-Department is a 1 to N (one to many).

Between Employee and Schedule is like that because 1 Employee has more Schedule that refers to him.

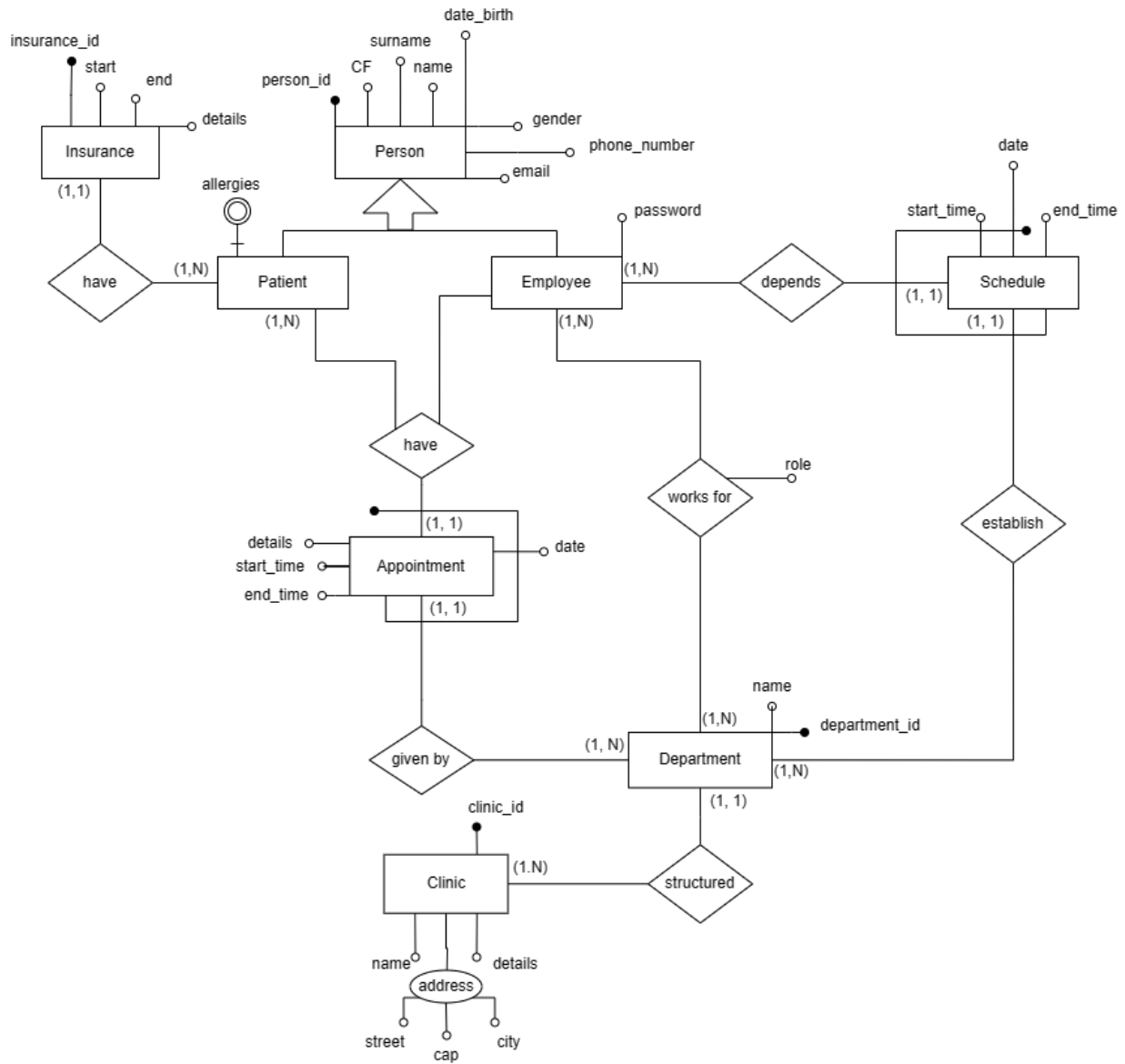
The Department and Schedule is as before, so 1 Department has more Schedules that are referred to the more Employees that work in it in a day.

Also, the cardinality between Insurance-Patient is 1 to many because, during the life, a Patient can have more Insurance.

External integrity constraint

1. More Insurance can be related to a Patient but not at the same time
2. The relations “works for” have role as indentificator because an employee can have different roles also in the same department.
3. A Department must have a Director that is defined by the role attribute of the relation “works for”; the director is only 1
4. For Schedule, an Employee must start and end the working day the same day; for example, start at 8.00 and end at 16.00 in the same day

ER Model



Insurance (insurance_id, patient, start, end, details)

key: insurance_id

foreignkey: Insurance[patient] $\subseteq$ Patient[person_id]

Person (person_id, CF, name, surname, date_birth, gender, phone_number, email)

key: person_id

Patient (patient_id, allergies)

key: patient_id

foreignkey: Patient[patient_id] $\subseteq$ Person[person_id]

constraint: allergies can be NULL

Employee (employee_id, password)

key: employee_id

foreignkey: Employee[employee_id] $\subseteq$ Person[person_id]

Department (department_id, name)

key: department_id

Clinic (clinic_id, name, street, cap, city, details)

key: clinic_id

Schedule (start_time, date, employee, department, end_time)

key: start_time, date

foreignkey: Schedule[employee] $\subseteq$ Employee [employee_id]

Schedule[department] $\subseteq$ Department[department_id]

Appointment (date, patient, employee, department, start_time, end_time, details)

key: date

foreignkey: Appointment[patient] $\sqsubseteq$ Patient[patient_id]

Appointment[employee] $\sqsubseteq$ Employee[employee_id]

Appointment[department] $\sqsubseteq$ Department[department_id]

Glossary of terms

TERM	DEFINITION
Insurance	A contract that requires your health insurer to pay some or all of Patient health care costs in exchange for a monthly payment.
Person	Is generic, defines both Patient and Employee common information
Patient	a Person receiving or registered to receive medical treatment
Employee	a person employed for wages or salary, in this case, in a Clinic; can have different roles
Schedule	As a time table, is a chart showing the working hours of each Employee
Clinic	an establishment or hospital where patients are given medical treatment or advice, especially of a specialist nature.
Appointment	an arrangement taken by a Patient to meet an Employee at a particular time and day
Department	Different part of a Clinic in which everyone have a specialisation

Data dictionary

Insurance

Column	Data Type	Description
insurance_id	VARCHAR(5)	PK of Insurance,
patient	VARCHAR(5)	FK of Insurance that refers to patient_id of Patient
start	DATE	Identifies when the insurance begin
end	DATE	Identifies when the insurance end
details	TEXT	description of other details

Person

Column	Data Type	Description
person_id	VARCHAR(5)	PK of every person (Patient or Employee)
CF	VARCHAR(10)	refers to the Fiscal Code of a person
name	VARCHAR(20)	The name of a person
surname	VARCHAR(20)	The surname of a person
date_birth	DATE	The date of birth
gender	CHAR(1)	Related to the gender, 'F' if is feminine of 'M' for male
phone_number	VARCHAR(10)	The phone number
email	VARCHAR(40)	The email of the patient

Patient

Column	Data Type	Description
patient_id	VARCHAR(5)	PK of patient, refers to the PK of Person
allergies	TEXT	all the possibly allergies of the Patient

Employee

Column	Data Type	Description
employee_id	VARCHAR(5)	PK of employee, refers to the PK of Person
password	VARCHAR(8)	Password used to access to the Clinic System by combining it with the personal email

Appointment

Column	Data Type	Description
date	DATE	The day of the appointment, with the patient_id,

		employee_id and department_id define the PK
patient	VARCHAR(5)	Is a FK that is referred to the PK of Patient
employee	VARCHAR(5)	Is a FK that refers to the PK of Employee
department	VARCHAR(5)	Is a FK that refers to the PK of Department
start_time	TIME	The time of the begin of the Appointment
end_time	TIME	The time of the end of the Appointment
details	TEXT	More details related with the Appointment

Department

Column	Data Type	Description
department_id	VARCHAR(5)	The PK of the Department
name	VARCHAR(30)	Defines the name of the department and so the possibly cure

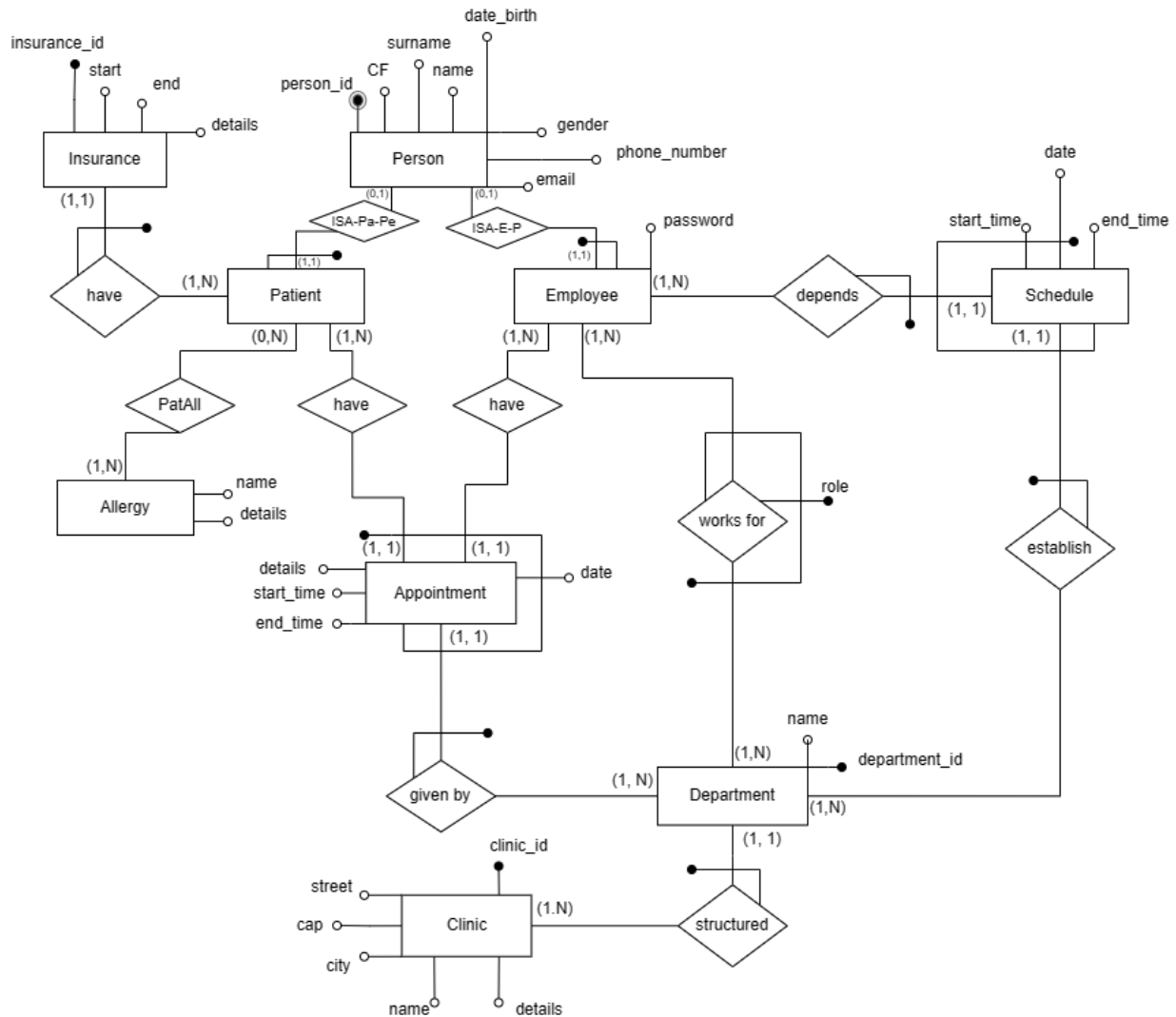
Clinic

Column	Data Type	Description
clinic_id	VARCHAR(5)	The PK of the Clinic
name	VARCHAR(20)	The name of the Clinic
street	VARCHAR(30)	The street address
cap	VARCHAR(5)	The cap of the city
city	VARCHAR(15)	The name of the city
details	TEXT	Other details related to the Clinic

Schedule

Column	Data Type	Description
start_time	TIME	PK, the begin of working hours
date	DATE	PK, the day of work
employee	VARCHAR(5)	FK that refers to the PK of the Employee
department	VARCHAR(5)	FK that refers to the PK of the Department
end_time	TIME	The end of working hours

Reconstructed ER



Person (person_id, CF, name, surname, date_birth, gender, phone_number, email)
 key: person_id

Patient (patient_id)
 key: patient_id
 foreignkey: Patient[patient_id] $\subseteq$ Person[person_id]
 constraint: allergies can be NULL

Allergy (name, patient, details)
 key: name

foreignkey: Allergy[patient] $\sqsubseteq$ Patient[patient_id]

Insurance (insurance_id, patient, start, end, details)

key: insurance_id

foreignkey: Insurance[patient] $\sqsubseteq$ Patient[patient_id]

Employee (employee_id, password)

key: employee_id

foreignkey: Employee[employee_id] $\sqsubseteq$ Person[person_id]

Clinic (clinic_id, name, street, cap, city, details)

key: clinic_id

Department (department_id, clinic, name)

key: department_id

foreignkey: Department[clinic] $\sqsubseteq$ Clinic[clinic_id]

Schedule (start_time, date, employee, department, end_time)

key: start_time, date

foreignkey: Schedule[employee] $\sqsubseteq$ Employee [employee_id]

Schedule[department] $\sqsubseteq$ Department[department_id]

Appointment (date, employee, patient, department, start_time, end_time, details)

key: date

foreignkey: Appointment[department] $\sqsubseteq$ Department[department_id]

Appointment[patient] $\sqsubseteq$ Patient[patient_id]

Appointment[employee] $\sqsubseteq$ Employee[employee_id]

WorkFor (employee, department, role)

key: role

foreignkey: WorkFor[employee] $\sqsubseteq$ Employee[employee_id]

WorkFor[department] $\sqsubseteq$ Department[department_id]

Reconstructed data dictionary

Insurance

Column	Data Type	Description
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insurance_id	VARCHAR(5)	PK of Insurance,
patient	VARCHAR(5)	FK of Insurance that refers to patient_id of Patient
start	DATE	Identifies when the insurance begin
end	DATE	Identifies when the insurance end
details	TEXT	description of other details

Person

Column	Data Type	Description
person_id	VARCHAR(5)	PK of every person (Patient or Employee)
CF	VARCHAR(10)	refers to the Fiscal Code of a person
name	VARCHAR(20)	The name of a person
surname	VARCHAR(20)	The surname of a person
date_birth	DATE	The date of birth
gender	CHAR(1)	Related to the gender, 'F' if is feminine of 'M' for male
phone_number	VARCHAR(10)	The phone number
email	VARCHAR(40)	The email of the patient

Patient

Column	Data Type	Description
patient_id	VARCHAR(5)	PK of patient, refers to the PK of Person

Allergy

Column	Data Type	Description
name	VARCHAR(5)	PK of Allergy
patient	VARCHAR(5)	FK of Allergy, refers to patient_id of Patient
details	TEXT	Details on the allergy

Employee

Column	Data Type	Description
employee_id	VARCHAR(5)	PK of employee, refers to the PK of Person
password	VARCHAR(8)	Password used to access to the Clinic System by

		combining it with the personal email
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Appointment

Column	Data Type	Description
date	DATE	The day of the appointment, with the patient_id, employee_id and department_id define the PK
department	VARCHAR(5)	Is a FK that refers to the PK of Department
patient	VARCHAR(5)	Is a FK that refers to the PK of Patient
employee	VARCHAR(5)	Is a FK that refers to the PK of Employee
start_time	TIME	The time of the begin of the Appointment
end_time	TIME	The time of the end of the Appointment
details	TEXT	More details related with the Appointment

Department

Column	Data Type	Description
department_id	VARCHAR(5)	The PK of the Department
clinic	VARCHAR(5)	FK of Department, refers to clinic_id of Clinic
name	VARCHAR(30)	Defines the name of the department and so the possibly cure

Clinic

Column	Data Type	Description
clinic_id	VARCHAR(5)	The PK of the Clinic
name	VARCHAR(20)	The name of the Clinic
street	VARCHAR(30)	The street address
cap	VARCHAR(5)	The cap of the city
city	VARCHAR(15)	The name of the city
details	TEXT	Other details related to the Clinic

Schedule

Column	Data Type	Description
start_time	TIME	PK, the begin of working hours
date	DATE	PK, the day of work

employee	VARCHAR(5)	FK that refers to the PK of the Employee
department	VARCHAR(5)	FK that refers to the PK of the Department
end_time	TIME	The end of working hours

WorkFor

Column	Data Type	Description
employee	VARCHAR(5)	FK that refers to the PK of the Employee
department	VARCHAR(5)	FK that refers to the PK of the Department
role	VARCHAR(30)	PK of the relation "WorkFor"

Table of Volume

```
CREATE TABLE Person (
    person_id VARCHAR(5) NOT NULL,
    CF VARCHAR(10) NOT NULL,
    name VARCHAR(20) NOT NULL DEFAULT "",
    surname VARCHAR(20) NOT NULL DEFAULT "",
    date_birth DATE NOT NULL,
    gender CHAR(1) NOT NULL DEFAULT 'F',
    phone_number VARCHAR(10) NOT NULL,
    email VARCHAR(40) NOT NULL,
    CONSTRAINT PrimaryKeyP PRIMARY KEY(person_id)
);
```

```
INSERT INTO Person(person_id, CF, name, surname, date_birth, gender, phone_number,
email) VALUES
('pT6r7', 'CF8765JH32', 'Jane', 'Doe', '1985-02-12', 'F', '5551234567', 'jane.doe@email.com'),
('pE8w1', 'CF6543KL98', 'John', 'Smith', '1978-07-25', 'M', '5552345678',
'john.smith@email.com'),
('pJ4x9', 'CF9087GF45', 'Mary', 'Johnson', '1990-11-08', 'F', '5553456789',
'mary.johnson@email.com'),
('pS3q8', 'CF2345DT98', 'David', 'Brown', '1982-05-18', 'M', '5554567890',
'david.brown@email.com'),
('pK6t2', 'CF7654YH21', 'Sarah', 'Lee', '1987-08-30', 'F', '5555678901', 'sarah.lee@email.com'),
('pF5z9', 'CF9876LK54', 'Michael', 'Davis', '1975-04-02', 'M', '5556789012',
'michael.davis@email.com'),
('pN2v6', 'CF5467FG23', 'Emily', 'Wilson', '1988-12-15', 'F', '5557890123',
'emily.wilson@email.com'),
('pG9p4', 'CF1234GT56', 'James', 'Taylor', '1980-09-27', 'M', '5558901234',
'james.taylor@email.com'),
('pR4m8', 'CF8976RF12', 'Jessica', 'Anderson', '1992-06-10', 'F', '5559012345',
'jessica.anderson@email.com'),
```

```
(('pB6k3', 'CF4567KM89', 'Robert', 'Martin', '1977-03-23', 'M', '5550123456',
'robert.martin@email.com'),
('pL7r6', 'CF7890DS43', 'Samantha', 'Thompson', '1995-01-05', 'F', '5551234567',
'samantha.thompson@email.com'),
('pC4t8', 'CF2345LR76', 'Daniel', 'White', '1984-10-17', 'M', '5552345678',
'daniel.white@email.com'),
('pD9r5', 'CF9087JH21', 'Amanda', 'Garcia', '1989-07-28', 'F', '5553456789',
'amanda.garcia@email.com'),
('pQ1k9', 'CF6543KL98', 'Christopher', 'Martinez', '1981-04-11', 'M', '5554567890',
'christopher.martinez@email.com'),
('pH4w1', 'CF8765PO43', 'Lauren', 'Hernandez', '1993-11-23', 'F', '5555678901',
'lauren.hernandez@email.com'),
('pT6r5', 'RBNMNN12A3', 'Alice', 'Smith', '1990-05-20', 'F', '1234567890',
'alice.smith@example.com'),
('pE8w3', 'DFGHJW98YU', 'Bob', 'Johnson', '1985-09-12', 'M', '2345678901',
'bob.johnson@example.com'),
('pJ4x2', 'KJHGF876TR', 'Carla', 'Garcia', '1995-01-02', 'F', '3456789012',
'carla.garcia@example.com'),
('pS3q1', 'BNMVCD456E', 'David', 'Brown', '1980-07-15', 'M', '4567890123',
'david.brown@example.com'),
('pK6t4', 'ZXCVCBN456G', 'Emily', 'Davis', '1992-03-25', 'F', '5678901234',
'emily.davis@example.com'),
('pF5z6', 'QWERTY123U', 'Frank', 'Lee', '1987-11-03', 'M', '6789012345',
'frank.lee@example.com'),
('pG9p7', 'POIUYT765R', 'Gina', 'Chen', '1998-02-18', 'F', '7890123456',
'gina.chen@example.com'),
('pR4m9', 'LKJHGF345E', 'Henry', 'Jackson', '1983-06-30', 'M', '8901234567',
'henry.jackson@example.com'),
('pL7r8', 'AZSXDC987V', 'Isabelle', 'Wilson', '1994-09-08', 'F', '9012345678',
'isabelle.wilson@example.com'),
('pC4t0', 'MNBVCX345R', 'John', 'Taylor', '1982-12-22', 'M', '0123456789',
'john.taylor@example.com'));
```

```
CREATE TABLE Patient (
    patient_id VARCHAR(5),
    CONSTRAINT PrimaryKeyPat PRIMARY KEY(patient_id),
    CONSTRAINT ForeignKeyPat FOREIGN KEY(patient_id) REFERENCES
    Person(person_id)
);
```

```
INSERT INTO Patient(patient_id) VALUES
('pT6r7'), ('pE8w1'), ('pJ4x9'), ('pS3q8'), ('pK6t2'), ('pF5z9'), ('pN2v6'), ('pG9p4'), ('pB6k3'),
('pR4m8'), ('pL7r6'), ('pC4t8'), ('pD9r5'), ('pQ1k9'), ('pH4w1');
```

```
CREATE TABLE Allergy (
    patient VARCHAR(5) NOT NULL,
    name VARCHAR(15) NOT NULL,
    details TEXT DEFAULT "",
    CONSTRAINT PrimaryKeyA PRIMARY KEY(patient),
    CONSTRAINT ForeignKeyA FOREIGN KEY(patient) REFERENCES Patient(patient_id)
);
```

```
INSERT INTO Allergy(patient, name, details) VALUES
('pT6r7', 'Pollen', 'Sneezing, runny nose, itchy eyes'),
('pE8w1', 'Peanuts', 'Hives, swelling, difficulty breathing'),
('pS3q8', 'Penicillin', 'Hives, rash, itching'),
('pK6t2', 'Shellfish', 'Nausea, vomiting, diarrhoea'),
('pG9p4', 'Latex', 'Skin rash, itching, hives'),
('pL7r6', 'Eggs', 'Hives, swelling, difficulty breathing'),
('pC4t8', 'Grass pollen', 'Sneezing, runny nose, itchy eyes'),
('pD9r5', 'Tree nuts', 'Swelling, itching, difficulty breathing'),
('pQ1k9', 'Aspirin', 'Hives, rash, difficulty breathing');
```

```
CREATE TABLE Insurance(
    insurance_id VARCHAR(5) NOT NULL,
    patient VARCHAR(5) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    details TEXT DEFAULT "",
    CONSTRAINT PrimaryKeyI PRIMARY KEY(insurance_id),
    CONSTRAINT ForeignKeyI FOREIGN KEY(patient) REFERENCES Patient(patient_id)
);
```

```
INSERT INTO Insurance(insurance_id, patient, start_date, end_date) VALUES
('iN3r6','pT6r7', '2022-01-01', '2023-01-01', 'Life Insurance'),
('iA2s7','pE8w1', '2022-03-15', '2023-03-15', 'Health Insurance'),
('iG1t4', 'pJ4x9', '2022-05-20', '2023-05-20', 'Health Insurance'),
('iM5w2', 'pS3q8', '2022-07-10', '2023-07-10', 'Life Insurance'),
('iF8x6', 'pK6t2', '2022-09-01', '2023-09-01', 'Health Insurance'),
('iD1f7', 'pF5z9', '2022-02-15', '2023-02-15', 'Health Insurance'),
('iL9t2', 'pN2v6', '2022-06-01', '2023-06-01', 'Life Insurance'),
('iR4g5', 'pG9p4', '2022-08-12', '2023-08-12', 'Health Insurance'),
('iE5h7', 'pR4m8', '2022-11-05', '2023-11-05', 'Life Insurance'),
('iX2t1', 'pB6k3', '2022-12-25', '2023-12-25', 'Health Insurance'),
('iW8z9', 'pL7r6', '2022-04-17', '2023-04-17', 'CHearthar Insurance'),
('iS9u2', 'pC4t8', '2022-10-01', '2023-10-01', 'Health Insurance'),
('iK3v7', 'pD9r5', '2022-05-30', '2023-05-30', 'Health Insurance'),
('iP5n6', 'pQ1k9', '2022-07-21', '2023-07-21', 'Life Insurance'),
('iB1d7', 'pH4w1', '2022-09-15', '2023-09-15', 'TrHealthavel Insurance');
```

```
CREATE TABLE Employee (  
    employee_id VARCHAR(5) NOT NULL,  
    password VARCHAR(8) NOT NULL,  
    CONSTRAINT PrimaryKeyEmp PRIMARY KEY(employee_id),  
    CONSTRAINT ForeignKeyEmp FOREIGN KEY(employee_id) REFERENCES  
    Person(person_id)  
);
```

```
INSERT INTO Employee(employee_id, password) VALUES  
( 'pJ4x9', 'uBp5wR7@'),  
( 'pF5z9', 'cNq2kG8@'),  
( 'pR4m8', 'eMg7fV9#'),  
( 'pT6r5', 'tKb6xZ3^'),  
( 'pE8w3', 'wPf9rC2%'),  
( 'pJ4x2', 'hLd1mF6&'),  
( 'pS3q1', 'yJn7bD5*'),  
( 'pK6t4', 'fGh4tE8$'),  
( 'pF5z6', 'iQl9oP2('),  
( 'pG9p7', 'sTj6eH8)'),  
( 'pR4m9', 'aXp3kN7@'),  
( 'pL7r8', 'xZv1mS4#'),  
( 'pC4t0', 'cBn8kF6^');
```

```
CREATE TABLE Clinic(  
    clinic_id VARCHAR(5) NOT NULL,  
    name VARCHAR(20) NOT NULL,  
    street VARCHAR(30) NOT NULL,  
    cap VARCHAR(5) NOT NULL,  
    city VARCHAR(15) NOT NULL,  
    details TEXT DEFAULT "",  
    CONSTRAINT PrimarykeyC PRIMARY KEY(clinic_id)  
);
```

```
INSERT INTO Clinic(clinic_id, name, street, cap, city, details) VALUES  
( 'C0001', 'ABC Clinic', '123 Main Street', '100', 'New York', 'Well-equipped facility'),  
( 'C0002', 'XYZ Medical Center', '456 Elm Avenue', '150', 'Los Angeles', 'Offers specialized  
services'),  
( 'C0003', 'HealthFirst', '789 Oak Drive', '80', 'Chicago', 'Open 24/7'),  
( 'C0004', 'City Clinic', '234 Maple Lane', '120', 'Boston', 'Multilingual staff'),  
( 'C0005', 'Coastal Care', '567 Pine Street', '90', 'Miami', 'Accepts insurance'),  
( 'C0006', 'Sunrise Hospital', '890 Cedar Avenue', '200', 'Las Vegas', 'Level I trauma center'),  
( 'C0007', 'Green Valley Medical', '1234 Walnut Boulevard', '130', 'Phoenix', 'On-site  
laboratory'),
```



```
('C0008', 'Wellness Clinic', '432 Cherry Lane', '70', 'Houston', 'Integrative medicine approach'),
('C0009', 'Cityscape Medical', '765 Spruce Road', '110', 'San Francisco', 'Specializes in sports medicine'),
('C0010', 'Harmony Health', '987 Birch Street', '180', 'Seattle', 'Offers alternative therapies');
```

```
CREATE TABLE Department(
    department_id VARCHAR(5) NOT NULL,
    clinic VARCHAR(5) NOT NULL,
    name VARCHAR(30) NOT NULL,
    CONSTRAINT PrimaryKeyD PRIMARY KEY(department_id),
    CONSTRAINT ForeignKeyDC FOREIGN KEY(clinic) REFERENCES Clinic(clinic_id)
);
```

```
INSERT INTO Department(department_id, clinic, name) VALUES
('D0001', 'C0001', 'General Medicine'),
('D0002', 'C0001', 'Pediatrics'),
('D0003', 'C0002', 'Cardiology'),
('D0004', 'C0002', 'Dermatology'),
('D0005', 'C0003', 'Orthopedics'),
('D0006', 'C0003', 'Ophthalmology'),
('D0007', 'C0004', 'ENT (Otolaryngology)'),
('D0008', 'C0004', 'Gastroenterology'),
('D0009', 'C0005', 'Neurology'),
('D0010', 'C0005', 'Urology'),
('D0011', 'C0006', 'Obstetrics/Gynecology'),
('D0012', 'C0006', 'Oncology'),
('D0013', 'C0007', 'Radiology'),
('D0014', 'C0007', 'Pulmonology'),
('D0015', 'C0008', 'Psychiatry'),
('D0016', 'C0008', 'Rehabilitation'),
('D0017', 'C0009', 'Endocrinology'),
('D0018', 'C0009', 'Hematology'),
('D0019', 'C0010', 'Allergy/Immunology'),
('D0020', 'C0010', 'Plastic Surgery'),
('D0021', 'C0001', 'Dental'),
('D0022', 'C0001', 'Psychiatry'),
('D0023', 'C0002', 'Internal Medicine'),
('D0024', 'C0002', 'Rheumatology');
```



```
('D0025', 'C0003', 'Pulmonology');
```

```
CREATE TABLE Schedule(
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    date DATE NOT NULL,
    employee VARCHAR(5) NOT NULL,
    department VARCHAR(5) NOT NULL,
    CONSTRAINT PrimaryKeyS PRIMARY KEY(start_time, date, employee, department),
    CONSTRAINT ForeignKeySE FOREIGN KEY(employee) REFERENCES
Employee(employee_id),
    CONSTRAINT ForeignKeySD FOREIGN KEY(department) REFERENCES
Department(department_id)
);
```

```
INSERT INTO Schedule(start_time, end_time, date, employee, department) VALUES
('11:00:00', '13:00:00', '2023-07-14', 'pG9p7', 'D0002'),
('13:30:00', '15:30:00', '2023-07-14', 'pR4m9', 'D0003'),
('16:00:00', '18:00:00', '2023-07-14', 'pL7r8', 'D0004'),
('08:00:00', '10:00:00', '2023-07-15', 'pC4t0', 'D0005'),
('10:30:00', '12:30:00', '2023-07-15', 'pJ4x9', 'D0006'),
('13:00:00', '15:00:00', '2023-07-15', 'pF5z9', 'D0007'),
('15:30:00', '17:30:00', '2023-07-15', 'pR4m8', 'D0008'),
('08:30:00', '10:30:00', '2023-07-16', 'pT6r5', 'D0009'),
('11:00:00', '13:00:00', '2023-07-16', 'pE8w3', 'D0010'),
('13:30:00', '15:30:00', '2023-07-16', 'pJ4x2', 'D0011'),
('16:00:00', '18:00:00', '2023-07-16', 'pS3q1', 'D0012'),
('09:00:00', '11:00:00', '2023-07-17', 'pK6t4', 'D0013'),
('11:30:00', '13:30:00', '2023-07-17', 'pF5z6', 'D0014'),
('14:00:00', '16:00:00', '2023-07-17', 'pG9p7', 'D0015'),
('16:30:00', '18:30:00', '2023-07-17', 'pR4m9', 'D0016'),
('08:00:00', '10:00:00', '2023-07-18', 'pL7r8', 'D0017'),
('10:30:00', '12:30:00', '2023-07-18', 'pC4t0', 'D0018'),
('13:00:00', '15:00:00', '2023-07-18', 'pJ4x9', 'D0019'),
('15:30:00', '17:30:00', '2023-07-18', 'pF5z9', 'D0020'),
```

```
(
    '08:30:00', '10:30:00', '2023-07-19', 'pR4m8', 'D0001'),
    ('11:00:00', '13:00:00', '2023-07-19', 'pT6r5', 'D0002'),
    ('13:30:00', '15:30:00', '2023-07-19', 'pE8w3', 'D0003'),
    ('16:00:00', '18:00:00', '2023-07-19', 'pJ4x2', 'D0004'),
    ('09:00:00', '11:00:00', '2023-07-20', 'pS3q1', 'D0005'),
    ('11:30:00', '13:30:00', '2023-07-20', 'pK6t4', 'D0006'),
    ('14:00:00', '16:00:00', '2023-07-20', 'pF5z6', 'D0007'),
    ('16:30:00', '18:30:00', '2023-07-20', 'pG9p7', 'D0008'),
    ('08:00:00', '10:00:00', '2023-07-21', 'pR4m9', 'D0009'),
    ('10:30:00', '12:30:00', '2023-07-21', 'pL7r8', 'D0010'),
    ('13:00:00', '15:00:00', '2023-07-21', 'pC4t0', 'D0011'),
    ('15:30:00', '17:30:00', '2023-07-21', 'pJ4x9', 'D0012'),
    ('08:30:00', '10:30:00', '2023-07-22', 'pF5z9', 'D0013'),
    ('11:00:00', '13:00:00', '2023-07-22', 'pG9p7', 'D0014'),
    ('13:30:00', '15:30:00', '2023-07-22', 'pR4m9', 'D0015'),
    ('16:00:00', '18:00:00', '2023-07-22', 'pL7r8', 'D0016'),
    ('08:00:00', '10:00:00', '2023-07-23', 'pC4t0', 'D0017'),
    ('10:30:00', '12:30:00', '2023-07-23', 'pJ4x9', 'D0018'),
    ('13:00:00', '15:00:00', '2023-07-23', 'pF5z9', 'D0019'),
    ('15:30:00', '17:30:00', '2023-07-23', 'pR4m8', 'D0020');

```

```
CREATE TABLE Appointment(
    date DATE NOT NULL,
    employee VARCHAR(5) NOT NULL,
    patient VARCHAR(5) NOT NULL,
    department VARCHAR(5) NOT NULL,
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    details TEXT DEFAULT "",
    CONSTRAINT PrimaryKeyApp PRIMARY KEY (date, employee, patient, department),
    CONSTRAINT ForeignKeyAE FOREIGN KEY(employee) REFERENCES
Employee(employee_id),
    CONSTRAINT ForeignKeyAD FOREIGN KEY(department) REFERENCES
Department(department_id),
    CONSTRAINT ForeignKeyAP FOREIGN KEY(patient) REFERENCES Patient(patient_id)
);

```

```
INSERT INTO Appointment(date, employee, patient, department, start_time, end_time,
details) VALUES
('2023-07-04', 'pC4t0', 'pB6k3', 'D0001', '10:00:00', '11:00:00', 'General check-up'),
('2023-07-04', 'pE8w3', 'pC4t8', 'D0002', '14:30:00', '15:30:00', 'Vaccine administration'),
('2023-07-04', 'pF5z6', 'pD9r5', 'D0003', '16:00:00', '17:00:00', 'EKG test'),
('2023-07-05', 'pF5z9', 'pE8w1', 'D0004', '09:30:00', '10:30:00', 'Skin biopsy'),
('2023-07-05', 'pG9p7', 'pF5z9', 'D0005', '11:30:00', '12:30:00', 'Knee examination'),
('2023-07-05', 'pJ4x2', 'pG9p4', 'D0006', '15:00:00', '16:00:00', 'Eye check-up'),
('2023-07-06', 'pJ4x9', 'pH4w1', 'D0007', '08:30:00', '09:30:00', 'X-ray of chest'),

```

```
(
('2023-07-06', 'pK6t4', 'pJ4x9', 'D0008', '10:30:00', '11:30:00', 'Stomach endoscopy'),
('2023-07-06', 'pL7r8', 'pK6t2', 'D0009', '14:00:00', '15:00:00', 'Nerve conduction study'),
('2023-07-07', 'pR4m8', 'pL7r6', 'D0010', '09:30:00', '10:30:00', 'Urinary tract ultrasound'),
('2023-07-07', 'pR4m9', 'pN2v6', 'D0011', '11:30:00', '12:30:00', 'Prenatal check-up'),
('2023-07-07', 'pS3q1', 'pQ1k9', 'D0012', '16:00:00', '17:00:00', 'Cancer treatment review'),
('2023-07-08', 'pT6r5', 'pR4m8', 'D0013', '08:30:00', '09:30:00', 'MRI scan'),
('2023-07-08', 'pC4t0', 'pS3q8', 'D0014', '11:30:00', '12:30:00', 'Lung function test'),
('2023-07-08', 'pE8w3', 'pT6r7', 'D0015', '15:00:00', '16:00:00', 'Mental health counseling'),
('2023-07-09', 'pF5z6', 'pB6k3', 'D0016', '10:30:00', '11:30:00', 'Physical therapy session'),
('2023-07-09', 'pF5z9', 'pC4t8', 'D0017', '13:30:00', '14:30:00', 'Diabetes management'),
('2023-07-09', 'pG9p7', 'pD9r5', 'D0018', '16:00:00', '17:00:00', 'Blood transfusion'),
('2023-07-10', 'pJ4x2', 'pE8w1', 'D0019', '08:30:00', '09:30:00', 'Allergy testing'),
('2023-07-10', 'pJ4x9', 'pF5z9', 'D0020', '11:30:00', '12:30:00', 'Cosmetic consultation'),
('2023-07-11', 'pK6t4', 'pG9p4', 'D0001', '09:00:00', '10:00:00', 'Tooth extraction'),
('2023-07-11', 'pL7r8', 'pQ1k9', 'D0002', '12:00:00', '13:00:00', 'Internal medicine check-up'),
('2023-07-11', 'pR4m9', 'pH4w1', 'D0003', '14:30:00', '15:30:00', 'Joint pain evaluation'),
('2023-07-12', 'pC4t0', 'pT6r7', 'D0004', '09:30:00', '10:30:00', 'Kidney ultrasound'),
('2023-07-12', 'pR4m8', 'pE8w1', 'D0005', '11:30:00', '12:30:00', 'Postnatal check-up'),
('2023-07-12', 'pS3q1', 'pJ4x9', 'D0006', '15:00:00', '16:00:00', 'Blood donation'),
('2023-07-13', 'pT6r5', 'pK6t2', 'D0007', '08:30:00', '09:30:00', 'Patch test'),
('2023-07-13', 'pC4t0', 'pL7r6', 'D0008', '10:30:00', '11:30:00', 'Botox treatment'),
('2023-07-13', 'pE8w3', 'pN2v6', 'D0009', '14:00:00', '15:00:00', 'Psychotherapy session'),
('2023-07-14', 'pF5z6', 'pQ1k9', 'D0010', '09:30:00', '10:30:00', 'Rehabilitation session'),
('2023-07-14', 'pF5z9', 'pR4m8', 'D0011', '11:30:00', '12:30:00', 'Diabetic foot care'),
('2023-07-14', 'pG9p7', 'pS3q8', 'D0012', '16:00:00', '17:00:00', 'Blood pressure monitoring'));
```

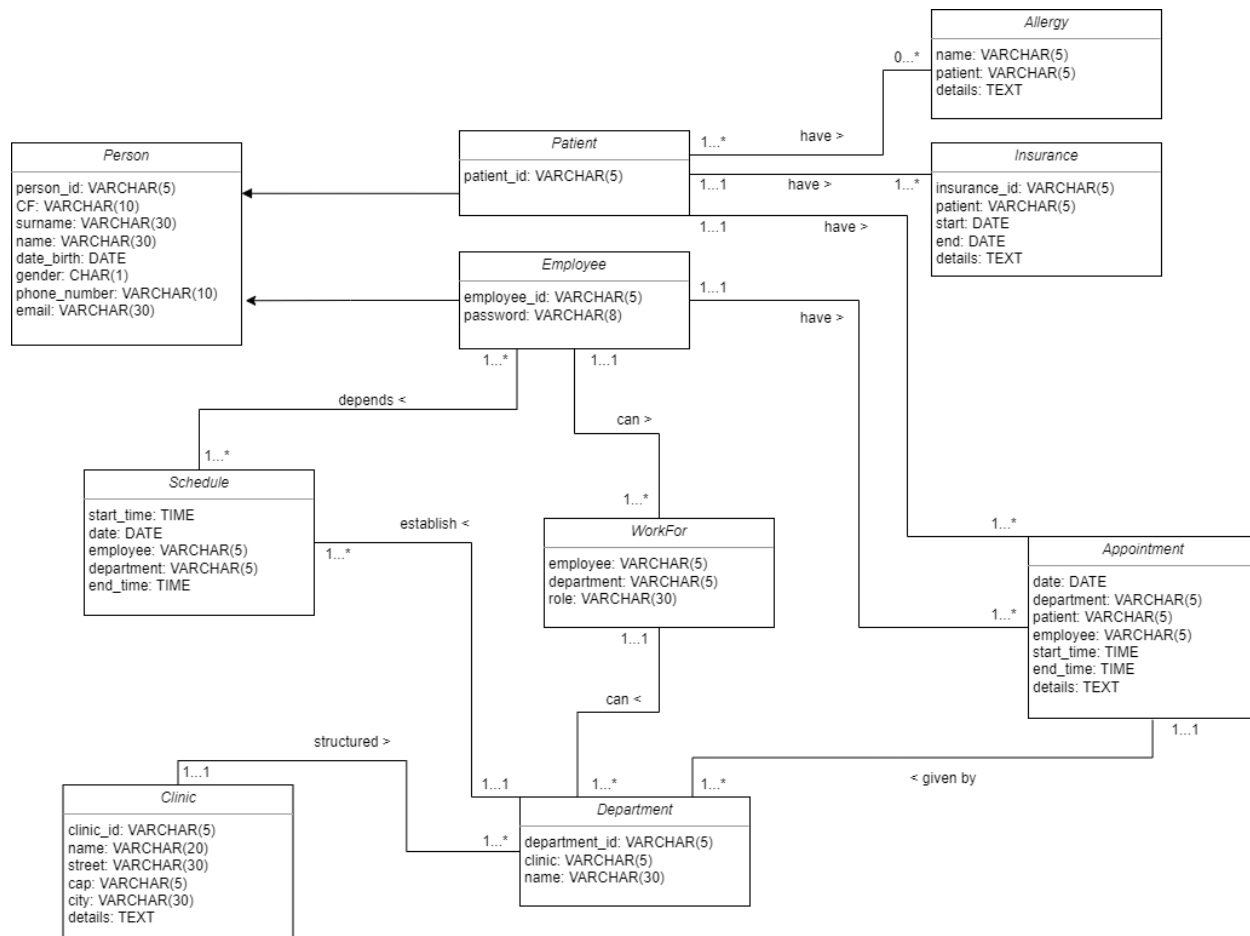
```
CREATE TABLE WorkFor(
    employee VARCHAR(5) NOT NULL,
    department VARCHAR(5) NOT NULL,
    role VARCHAR(30) NOT NULL,
    CONSTRAINT PrimaryKeyWF PRIMARY KEY(employee, department, role),
    CONSTRAINT ForeignKeyWFE FOREIGN KEY(employee) REFERENCES
Employee(employee_id),
    CONSTRAINT ForeignKeyWFD FOREIGN KEY(department) REFERENCES
Department(department_id)
);
```

```
INSERT INTO WorkFor(employee, department, role) VALUES
('pC4t0', 'D0001', 'Senior Physician'),
('pE8w3', 'D0002', 'Clinical Researcher'),
('pF5z6', 'D0003', 'Medical Technologist'),
('pF5z9', 'D0004', 'Pediatrician'),
('pG9p7', 'D0005', 'Cardiologist'),
('pJ4x2', 'D0006', 'Radiologist'),
('pJ4x9', 'D0007', 'Surgeon'),
('pK6t4', 'D0008', 'Anesthesiologist'),
```



('pL7r8', 'D0009', 'Neurologist'),
('pR4m8', 'D0010', 'Psychiatrist'),
('pR4m9', 'D0011', 'Oncologist'),
('pS3q1', 'D0012', 'Gynecologist'),
('pT6r5', 'D0013', 'Orthopedic Surgeon'),
('pC4t0', 'D0014', 'Ophthalmologist'),
('pE8w3', 'D0015', 'ENT Specialist'),
('pF5z6', 'D0016', 'Dermatologist'),
('pF5z9', 'D0017', 'Endocrinologist'),
('pG9p7', 'D0018', 'Urologist'),
('pJ4x2', 'D0019', 'Nephrologist'),
('pJ4x9', 'D0020', 'Gastroenterologist'),
('pK6t4', 'D0021', 'Pulmonologist'),
('pL7r8', 'D0022', 'Hematologist'),
('pR4m8', 'D0023', 'Rheumatologist'),
('pR4m9', 'D0024', 'Allergist'),
('pS3q1', 'D0025', 'Family Physician'),
('pT6r5', 'D0006', 'General Practitioner'),
('pC4t0', 'D0007', 'Emergency Physician'),
('pE8w3', 'D0008', 'Pathologist'),
('pF5z6', 'D0009', 'Medical Geneticist'),
('pF5z9', 'D0010', 'Physical Therapist'),
('pG9p7', 'D0011', 'Occupational Therapist');

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